

Auteur: Alhassane Balde
Studierichting: Informatica
Oefeningenreeks 2E nr. 5.8

$$\begin{aligned} & \lim_{x \rightarrow 0} \sqrt[x]{\frac{1+x}{1-x}} \\ &= \lim_{x \rightarrow 0} \left(\frac{1+x}{1-x} \right)^{\frac{1}{x}} \\ &= \lim_{t \rightarrow \infty} \left(\frac{1 + \frac{1}{t}}{1 - \frac{1}{t}} \right)^t \\ &= \frac{\lim_{t \rightarrow \infty} \left(1 + \frac{1}{t} \right)^t}{\lim_{t \rightarrow \infty} \left(1 - \frac{1}{t} \right)^t} \\ &= \frac{e}{e^{-1}} = e \cdot e = e^2 \end{aligned}$$

De berekening van $\lim_{t \rightarrow \infty} \left(1 - \frac{1}{t} \right)^t$ staat in oefening 4.9.

References